GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 02



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Calendar

100	5
200	3
300	1
400	0

ODD DAYS

100 yrs ⇒ 5 odd days

- L.Y → 24 → 48 → 0
- N.Y → 76 → 76 → 0

0 + 0 = 12

Topics to be Covered



Topic-1 Understanding Clock

Topic-2 Clock concept through Relative Speed

Topic-3 Questionnaire

CALENDAR REPETITION

2016 ✓

L.Y

2016 → 2

2017 → 1

2018 → 1

2019 → 1

2020 → 2

2021 N.Y

2044 ✓

Conditional Question

2016 ✓
+ 28
2044 ✓

2017
+ 6
2023

2012 ✓
+ 28
2040 ✓

L.Y. ⇒ 28 years
N.Y ⇒ 6 years
N.Y ⇒ 11 yrs
N.Y ⇒ 11 yrs
L.Y. ⇒ 28 yrs

2024 ✓
+ 28
2052 ✓

Conditional Question

400 days

100 ⇒ 5 Fri

200 ⇒ 3 Wed

300 ⇒ 1 Mon

400 ⇒ 0 Sun

9)400(44 week 9

36

40

36

4 → odd days

[MCQ]

#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

Saturday

1868 ⇒ 4 +2 = 6

1800 ⇒ 3

Assignment

A Monday

B Thursday

C Tuesday

D Saturday

J-3 A-3

F-0 S-2

M-3 O-2

A-2

M-3

J-2

O-3

68 ⇒ 1

L.Y N.Y

17 51

24 51

6 + 2 = 8

04

X17

68

[MCQ]

#Q. The day of the March 16th of any year is the same day of the week as the corresponding date in which month of the same year?

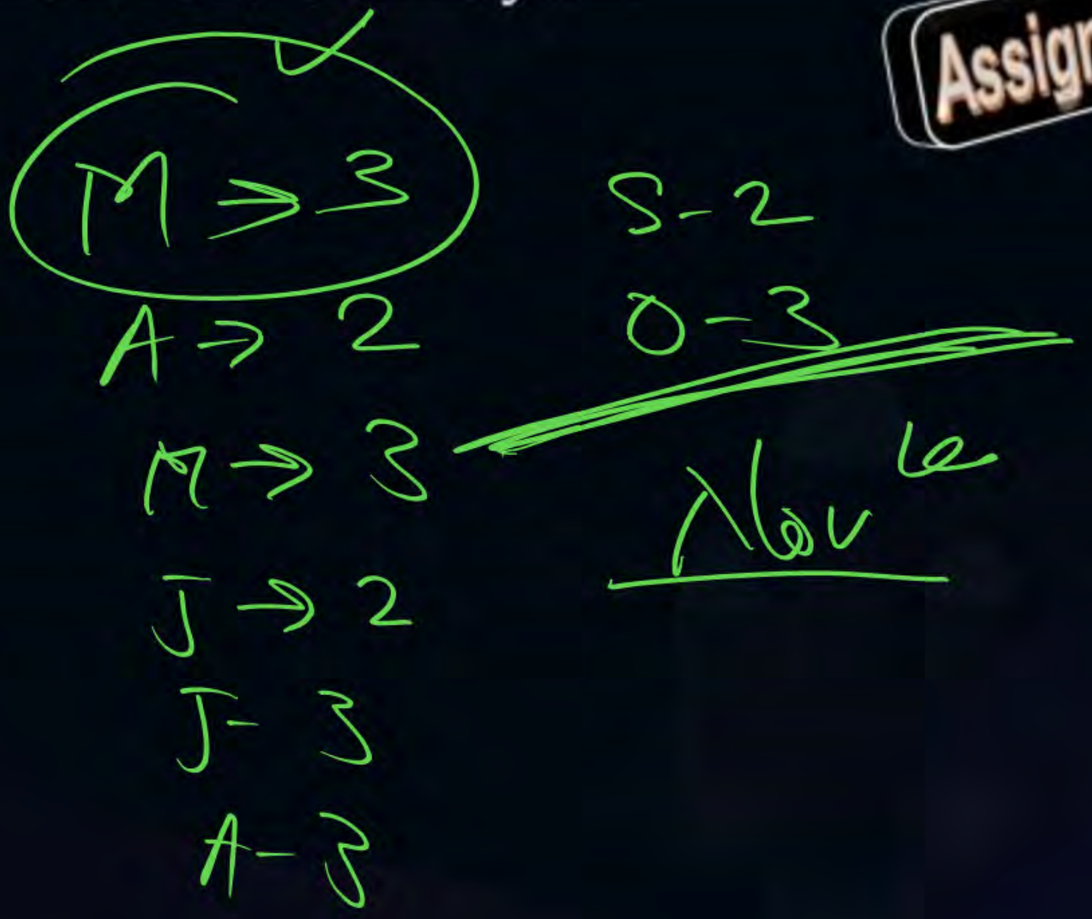
Assignment

A July

B November

C April

D September



[MCQ]

#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

+0

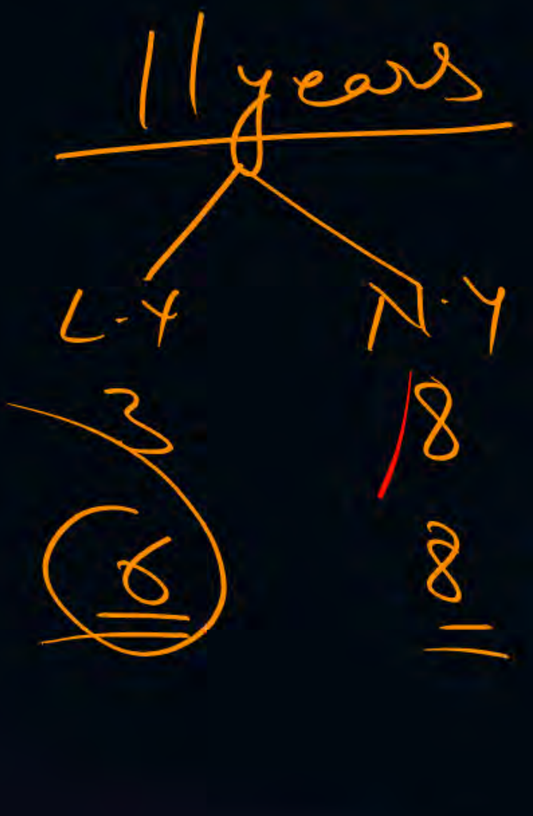
Assignment

A Sunday

B Wednesday

C Thursday

D Friday



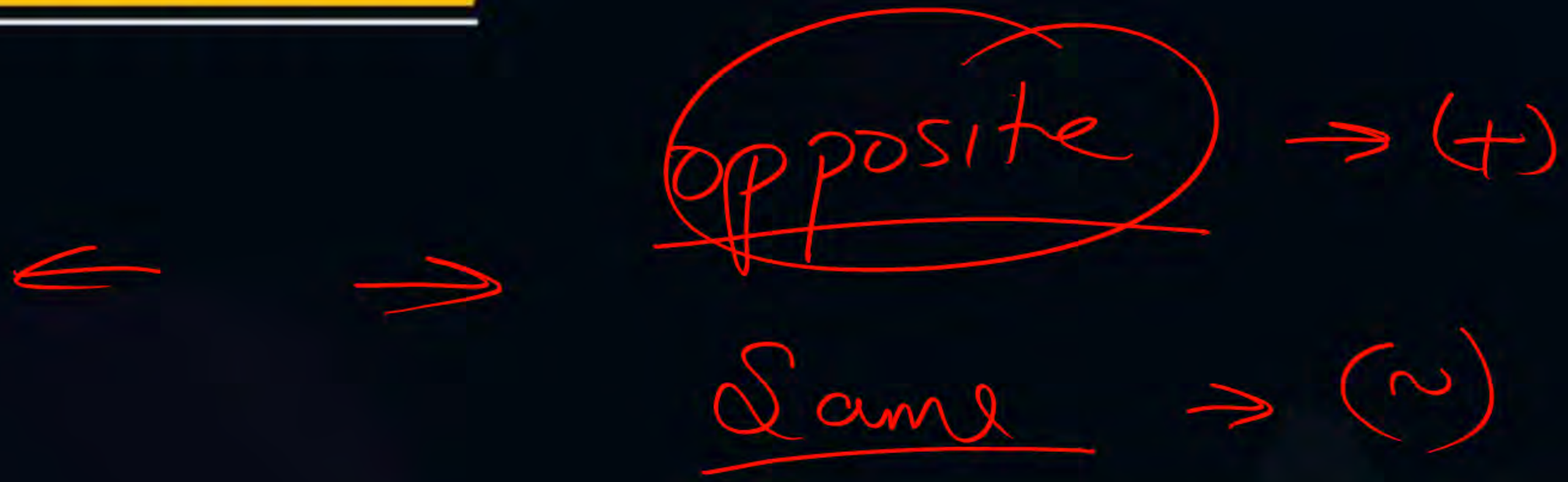
7 → 0 odd day

CLOCK



Relative Speed

Relative Speed:

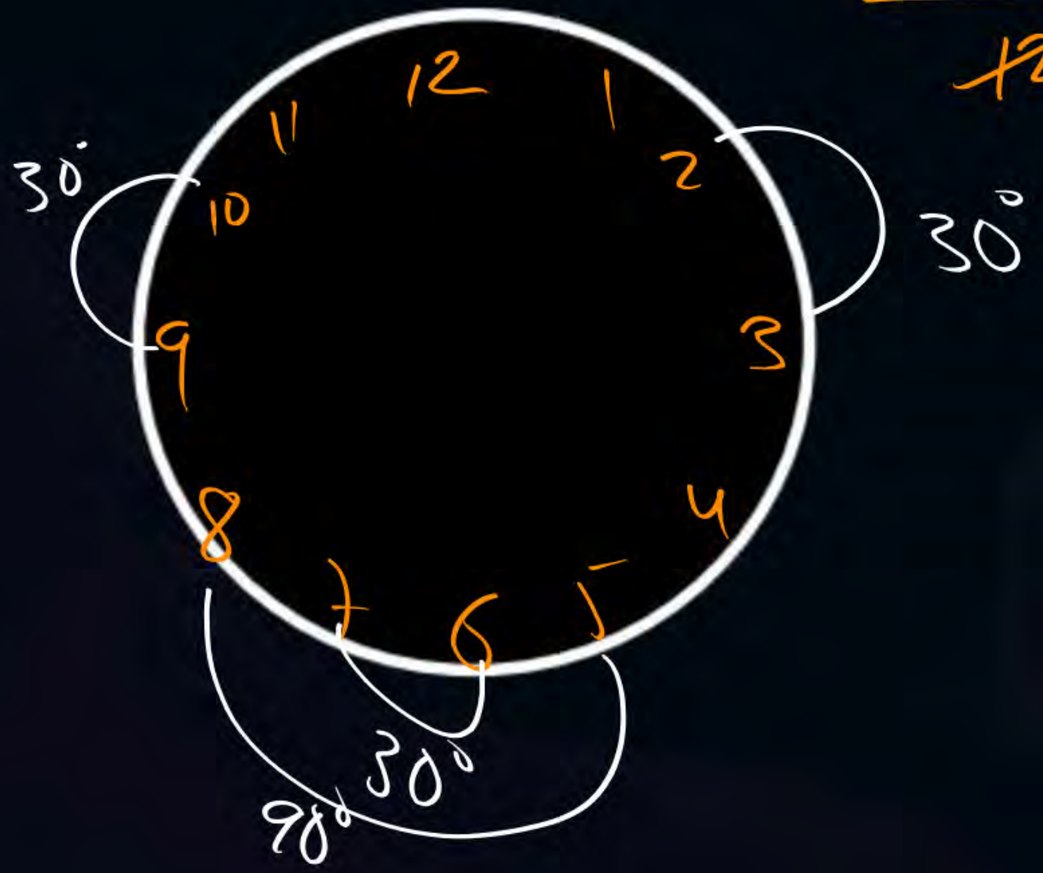


Relative Speed:

1. Hands — 2 — Hour Hand, Minute Hand
2. Movement — Same (clockwise)
3. Relative Speed — (−)

Dial of Clock

$$\frac{360^\circ}{12} = 30^\circ$$



Dial of Clock

M.H.

$= \frac{30^\circ}{5} = 6^\circ/\text{min}$



H.H.

$= \frac{30^\circ}{60}$

$= \frac{1}{2}^\circ/\text{min}$

$= 0.5^\circ/\text{min}$

To be Noted:

- ✓ Minute Hand Covers 30° in 5 minutes 30°/5 = 6°/min.
- ✓ Hour Hand Covers 30° in 60 minutes 30°/60 = ½ ° / min or 0.5°/min

Relative Speed

5.5°/min

Different Patterns of Questions:

1). Time ⇒ Angle?

2) Angle ⇒ Time?

3) Gain OR Loss

First Pattern:

Time → Angle?

6:00 → 180°

2:00 → 60°

7:00 → 210°

5:00 → 150°



Random time given:

1 : 20 → 80°

Difference

1 ⇒ 30°

20 × 5.5 ⇒ 110°

80°

80°

9 : 17 — 176.5°

9 ⇒ 270°

17 × 5.5 ⇒ 93.5°

176.5

Questionnaire:

345°

5:30

150°

5 → 150°

30 × 5.5 → 165°

15°



Questionnaire:

3:15 → 7.5°

3 ⇒ 90°

15 × 5.5 ⇒ 82.5°

7.5°

3:52.5



Questionnaire:



4:40

100°

4 ⇒ 120°

40 × 5.5 ⇒ 220°

100

260°



[MCQ]

#Q. At 9:45, the two hands of a clock make an angle of?

A 60°

B 45°

C 33 ⅓°

D 22 ½°

$9 \rightarrow 270^\circ$

$45 \times 5.5 \rightarrow 247.5^\circ$

22.5°

337.5°

Two Answers:

9:45

360
22.5
337.5

9 → 270°

45 × 5.5 → 247.5°

22.5°



Second Pattern:

Angle $\overset{u}{\rightarrow}$ Time?

0° OR Coincide:

12 hrs = 11 times

24 hrs = 22 times



180° OR Opposite:

0° & 180° Except

12 hrs = 11 times

24 hrs = 22 times



90° OR Right angle:

Any angle except 0° & 180°

12 hrs = 22 times

24 hrs = 44 times



Questionnaire:

$2:27\frac{3}{11}$

#Q. In between 2 O' clock and 3 O' clock at what time the hands of clock form 90°?

$$\frac{150^\circ \times 2}{5.5 \times 2} = \frac{300}{11}$$

$$= 27\frac{3}{11} \text{ min}$$



Questionnaire:

$4:03\frac{7}{11}$

$4:40$

#Q. In between 4 0' clock and 5 0' clock at what time the hands of clock form 100°?

$$\frac{20}{5.5} = \frac{40}{11} = 3\frac{7}{11} \text{ min}$$



$$\frac{220^\circ}{5.5} = \frac{440}{11} = 40 \text{ min}$$

Questionnaire:

Assignment

#Q. In between 6 0’ clock and 7 0’ clock at what time the hands of clock form 60°?



Questionnaire:

Assignment

#Q. In between 1 O' clock and 2 O' clock at what time the hands of clock form 100°?





2 mins Summary

Topic Clock

2 Answers

Time ⇒ Angle?

Angle ⇒ Time?

THANK - YOU